Vansh Bhasin
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## Education

**York University** **Fall 2023–Summer 2028 (Expected)**
*Bachelor of Science (Hons) in Computer Science* **GPA**: 8.6 / 9 *Toronto, ON*
**Relevant Coursework:** Object-Oriented Programming (OOP), Data Structures, Design and Analysis of Algorithms, Software Tools and Scripting, Building E-Commerce Systems, Introduction to Database Systems, Data Mining, Technical Countermeasures to Financial Crimes, Introduction to Computer Security

## Work Experience

**RBC** **May 2026–Present**
*Data Analyst Co-op* *Toronto, ON*
- Built a custom job scheduling platform integrated with the team's virtual machine and its task scheduler, automating **15+ ETL jobs** for team members to run off-hours and saving **5–6+ hours per day** of manual run time while freeing daytime compute.
- Launched a live dashboard displaying the real-time status of all scheduled runs, giving the team at-a-glance visibility into ETL pipeline health.
- Delivered a custom data-source connector **MCP (Model Context Protocol)** server and an "Ask Me Anything" AI agent on top of it, enabling non-technical teams to ask questions in plain English and query data without any SQL knowledge.
- Designed Tableau talent dashboards for executive-level reporting and cleaned, validated, and transformed large-scale HR and operational datasets using Python and SQL for RBC's Operations Process Engineering team.

**Sarit Micromobility, York University** **Jan 2026–Apr 2026**
*Research Assistant (Mobility Systems & Software Integration)* *Toronto, ON*
- Developed Python-based telemetry and data processing pipelines for next-generation micro-mobility vehicles, enabling real-time system monitoring and improving reliability across hardware and software layers.
- Engineered an RF and cloud-based remote lock/unlock system on Raspberry Pi, interfacing with electrical circuits and mechanical components to deliver secure, real-time vehicle access control for Sarit autonomous platforms.
- Integrated safety and compliance modules into the core mobility platform and built, tested, and documented embedded software components.

**Elder Lab, York University** **May 2025–Sep 2025**
*Research Assistant (Computer Vision & AI)* *Toronto, ON*
- Conducted drone-based traffic analysis by collecting aerial video data and deploying YOLO models for multi-class vehicle detection; streamlined drone calibration and model tuning using Agile project management, improving classification accuracy and reducing calibration time by **30%**.
- Utilized CVAT with Docker, Python (OpenCV, NumPy, Matplotlib, Pandas), FFmpeg, and Bash scripting for semi-automatic ground truthing, annotation, and preprocessing of large-scale image datasets, accelerating data labeling and preparation for deep learning workflows.
- Developed and maintained scalable computer vision pipelines for object detection and tracking in Linux, UNIX, and hybrid cloud environments using Git, Bash, and CI/CD tools, improving deployment speed by **40%** and enabling reproducible workflows across **3+ active research projects with GPU (CUDA)** acceleration.

**NGWN Lab, York University** **May 2024–Apr 2025**
*Research Assistant (Deep Learning & Data Analytics)* *Toronto, ON*
- Researched EMF exposure prediction using deep learning and generative data augmentation; published findings on 5G network EMF mapping.
- Created data pipelines connecting Python, SQL, and Power BI to support model validation, boosting validation workflows by **15%**.

## Projects

**Kronos** | *Python, Flask, PyTorch, Hugging Face, Heroku* **2026**
- Built a Flask market-foresight web app that runs **24–48 hour forecasts** on any Yahoo Finance ticker with the Kronos foundation model (mini, small, and base) loaded from Hugging Face checkpoints, alongside continuous OHLCV charts from 1D to 5Y.
- Wired prediction and OHLCV APIs with ticker search, tuned forecast defaults via **walk-forward backtests**, and deployed to Heroku (Python 3.12, CPU PyTorch) on a custom domain with a mobile-ready dark/light chart UI.

**ConsultHub** | *Java, Spring Boot, PostgreSQL, React, Gemini API, Docker* **2026**
- Architected a full-stack consulting booking REST API with Spring Boot, PostgreSQL, and JWT security, using the State design pattern to model a 6-state booking lifecycle and eliminate conditional transition logic across 15+ endpoints.
- Integrated a Gemini-powered AI customer assistant using **function calling (tool use)** to dynamically inject platform context such as services, availability, and policies into the model at runtime, delivering accurate recommendations without any model retraining or fine-tuning.
- Applied Strategy and Factory design patterns to build an extensible payment processing module supporting 4 payment methods (credit, debit, PayPal, bank transfer); containerized the full stack using Docker Compose with isolated PostgreSQL, H2 test, and Neon cloud databases.

**VisionCalc** | *React, TypeScript, MediaPipe, Computer Vision* **2026**
- Shipped a backend-free React + TypeScript gesture calculator that tracks hands in the browser with Google **MediaPipe HandLandmarker**, mapping extended-finger counts (0–5) to digits and distinct hand signals to +, −, ×, ÷, =, and clear.
- Isolated recognition into pure, **unit-tested** logic with ~0.9 s hold-to-confirm stabilization, a safe expression evaluator, and a no-webcam demo mode.

## Leadership and Volunteering

**VP of Operations** (AI and Data Science Society, York University) – Spearheaded logistics, event planning, and operational strategy for 15+ AI and data science initiatives, driving engagement and participation across 200+ students.
**Library Facilities Lead Hand** (York University Library, Sep 2024–Present) – Led staff coordination and special library projects to keep facility operations efficient, and supported patrons through assistance, policy communication, and wayfinding guidance.
**Peer Tutor** (Excel Lassonde) – Provided academic tutoring and mentoring in computer science and math courses, helping 30+ peers improve understanding and boosting average academic performance.

## Technical Skills

**Languages:** Python, Java, SQL, JavaScript/TypeScript, Bash/Shell, C++, R
**Frameworks & Libraries:** React, Spring Boot, Flask, Node.js, Express, Next.js, Tailwind CSS, Django, Angular, Vue.js, SwiftUI, JUnit, Cypress
**ML & Data:** PyTorch, TensorFlow, OpenCV, MediaPipe, scikit-learn, Pandas, NumPy, Spark, Hadoop, Hive, Databricks, MCP
**Cloud & DevOps:** Docker, AWS (Lambda, CloudFormation, CloudWatch), GCP, Azure, Kubernetes, Terraform, Helm, Ansible, Jenkins, CI/CD, Heroku
**Databases & BI:** PostgreSQL, MySQL, MongoDB, Oracle, Snowflake, Tableau, Power BI
**Tools & Practices:** Git, Bitbucket, Jupyter, Jira, Confluence, Agile